

Java toString() Method - Explanation & Examples

The `toString()` method in Java is used to convert an object into a string representation. It is defined in the `Object` class, which is the superclass of all Java classes.

Key Points About `toString()`

1. **Defined in `java.lang.Object`** – Every class in Java inherits this method.
2. **Default Behavior** – If not overridden, it returns:

```
ClassName@HashCode
```

Example:

```
MyClass@1a2b3c4d
```

3. **Commonly Overridden** – Custom classes should override `toString()` to provide meaningful string representations.

1. Default `toString()` Implementation

If you don't override `toString()`, Java uses the default implementation from `Object`:

```
java
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class Example {}

public class ToStringExample {
    public static void main(String[] args) {
        Example obj = new Example();
        System.out.println(obj.toString()); // Default toString()
    }
}
```

Output (Example)

```
css
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Example@7a81197d
```

This output contains:

- `Example` → Class name
 - `7a81197d` → Hash code in hexadecimal
-

2. Overriding `toString()` in a Custom Class

To get a meaningful string output, override `toString()` in your class.

Java code:

```
class Person {
    String name;
    int age;

    // Constructor
    Person(String name, int age) {
        this.name = name;
        this.age = age;
    }

    // Overriding toString() method
    @Override
    public String toString() {
        return "Person{name='" + name + "', age=" + age + "}";
    }
}

public class ToStringOverrideExample {
    public static void main(String[] args) {
        Person p1 = new Person("Alice", 25);
        System.out.println(p1.toString()); // Calls overridden toString()
    }
}
```

Output

```
Person{name='Alice', age=25}
```

This provides a **human-readable** representation instead of the default `ClassName@HashCode`.

3. Using `toString()` Implicitly

If you print an object, Java automatically calls `toString()`.

```
java
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public class ImplicitToString {
    public static void main(String[] args) {
        Person p = new Person("Bob", 30);
        System.out.println(p); // Implicitly calls toString()
    }
}
```

Output

```
Person{name='Bob', age=30}
```

Java **automatically** calls `toString()` when an object is printed.

4. `toString()` in Built-in Java Classes

Many Java classes override `toString()` for meaningful output.

Example: `ArrayList`

```
java
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import java.util.ArrayList;

public class ArrayListToString {
    public static void main(String[] args) {
        ArrayList<String> list = new ArrayList<>();
        list.add("Apple");
        list.add("Banana");
        System.out.println(list.toString()); // Calls overridden
toString()
    }
}
```

Output

```
csharp
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[Apple, Banana]
```

The `ArrayList` class overrides `toString()` to display elements in a readable format.

5. `toString()` with `Objects` Utility Class

Since Java 7, the `Objects` class provides a **null-safe** `toString()` method.

Example

```
java
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import java.util.Objects;

public class ObjectsToStringExample {
    public static void main(String[] args) {
        String str = null;
        System.out.println(Objects.toString(str, "Default Value")); //
Handles null safely
    }
}
```

Output

Default Value

If `str` is `null`, it prints "Default Value" instead of throwing `NullPointerException`.

6. `toString()` in Record Classes (Java 14+)

Java 14 introduced **records**, which automatically generate a `toString()` method.

```
java
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record Car(String brand, int year) {}

public class RecordToString {
    public static void main(String[] args) {
        Car car = new Car("Tesla", 2024);
        System.out.println(car);
    }
}
```

Output

```
Car[brand=Tesla, year=2024]
```

Records **auto-generate** `toString()`, so no need to override it manually.

Key Takeaways

- ✅ **Default `toString()`** prints `ClassName@HashCode`.
- ✅ **Overriding `toString()`** provides meaningful output.
- ✅ **Implicit calling** when printing an object.
- ✅ **Built-in Java classes** override `toString()`.
- ✅ Use `Objects.toString()` for null safety.
- ✅ **Records (Java 14+)** automatically generate `toString()`.